

PRAGMATIC REWARDS: INTRINSIC COMMUNICATIVE CLARITY IN MULTI-AGENT REINFORCEMENT LEARNING

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ABSTRACT

Cooperative RL agents often fail not because they act badly but because their intentions are illegible to their partners. Building on the structural correspondence between Maximum-Entropy RL and the Rational Speech Act (RSA) model of pragmatic communication, we import the RSA listener value—the log-posterior a listener assigns to an agent’s private state given its behavior—as an intrinsic reward for legibility, computed offline by a Pragmatic Reward Module that adds no latency at execution. The framework’s decisive design choice is the literal listener L_0 anchoring the reward. We prove that any instantaneous listener yields a variational lower bound on the intent–behavior mutual information—maximum-likelihood training merely tightens it—and give a gradient-saturation bound: a listener that decodes with high probability under the policy’s occupancy exerts a correspondingly small shaping gradient. Empirically, on scout-support, a benchmark whose oracle-gap criterion certifies that communication has task value, crude motion-grounded listeners close 60–82% of the oracle premium across three task variants with frozen hyperparameters (Welch $t \geq 4$), and a blind-partner control eliminates the gain while sparing the oracle: the effect is communication, not shaping. Learned listeners instead fail by succeeding—decoding any behavior, saturating, and shaping nothing—and we show this failure survives bounding the listener to the audience’s own viewpoint. RSA’s contrastive recursion is what rescues them: at a raised reward weight, the recursed learned listener recovers about a third of the premium (35% pooled over sixteen seeds including a pre-registered confirmation; $t = 3.4$, below our conservative headline bar and labeled accordingly) while its weight-matched literal control falls below baseline—recursion versus decoding alone is worth $t \approx 3.9$. What the hand-crafted winners still hold over the best learned listener is that they credit only overt signals a teammate can actually read. Sweeping partner speed and signal cost yields an empirical regularity—gain $\approx 0.5 \times$ premium where a premium exists—with four characterized failure modes, and we document pre-convergence evaluation hazards specific to co-adaptive communication measures.

Index Terms— Reinforcement Learning, Intrinsic Motivation, Multi-Agent Systems, Computational Pragmatics, Legibility

1. INTRODUCTION

Human collaboration is fundamentally a goal-oriented, strategic activity. To succeed, we do not just act to achieve our own goals; we act in ways that allow our partners to infer our beliefs and intentions, enabling better coordination. The same is true for artificial agents. In cooperative multi-agent reinforcement learning (MARL) environments, an agent’s success depends not only on its own actions but also on being understood by its teammates. Consider a cooperative navigation task in which each agent must cover a target of a preferred type: an agent that simply rushes toward its target may take a path that is ambiguous from a teammate’s perspective, causing the teammate to abandon a target the agent never intended to claim. A more sophisticated agent might take a slightly suboptimal path that unambiguously singles out its intended target, making its intention clear and enabling better team coordination.

This property of being easily understood is called “legibility” [1]. However, standard RL agents, optimized solely for an extrinsic task reward, have no incentive to be legible. This leads to our central question: how can we formally encourage an RL agent to choose actions that are not just extrinsically optimal for the task, but also intrinsically optimal for communicating its private state or intentions?

This work proposes a solution in the form of an intrinsic reward, R_{comm} , that quantifies the communicative clarity of an agent’s actions. To calculate this reward, we model an “observer” (e.g., another agent) who uses pragmatic inference to reason about why the agent chose a particular action over the available alternatives. We find the ideal formalism for this observer in the Rational Speech Act (RSA) framework [2], which provides a formal account of reasoning about communicative intent.

Any such observer model must be anchored in a literal listener L_0 : the assumed common ground that specifies how behavior would be interpreted in the absence

of pragmatic reasoning. Our analysis and experiments show that this choice is not an implementation detail but the decisive factor in whether pragmatic rewards help or hurt—and that it fails at both extremes. A hand-crafted listener can be misaligned with what the policy can express, incentivizing costly signaling that no observer can decode; a listener learned from the agents’ own behavior guarantees decodability by construction, but trained to convergence it decodes everything and demands nothing—its reward saturates and the shaping gradient provably vanishes. Mapping the productive region between these extremes—which listener, bounded how—is the empirical core of this paper.

Our contributions are as follows:

- We exhibit a structural correspondence between the objective functions of Max-Ent RL and the RSA framework—both are softmax choice models, identical in the single-step case under a temperature re-indexing—which motivates importing the RSA “listener value” as an intrinsic reward for legibility; purely communicative models such as CRSA [3] then correspond to the myopic, task-free operating point ($\gamma = 0$, $R_{\text{ext}} = 0$) of the same composite objective.
- We operationalize the theory as a Pragmatic Reward Module (PRM): an offline component, compatible with centralized training and decentralized execution (CTDE), that computes R_{comm} from replayed transitions and adds no latency at execution time.
- We characterize the space of literal listeners and prove that any listener makes the expected pragmatic reward a variational lower bound on the mutual information $I(M; S, A)$ between private intentions and observable behavior—maximum-likelihood training distinguishes the learned listener only by making the bound tightest. We complement this with a gradient-saturation result: when the listener decodes (nearly) perfectly on the policy’s support, the shaping gradient of the pragmatic reward (nearly) vanishes. Together these connect pragmatic rewards to information-regularization [4] and skill-discriminability [5] objectives, and explain why the tightest bound can be the worst reward.
- We map the boundary of the method by sweeping what early intent is worth and what signaling costs, obtaining an empirical regularity within this environment family—gain $\approx 0.5 \times$ premium across three environments, three partner speeds and two signal costs (Sec. 6.2 reports the fit with and without the failure-mode settings, with uncertainty)—

together with four distinct failure modes (no premium, no channel, partner too slow to act, signal too costly) that make “when pragmatism pays” a checkable condition rather than an empirical accident.

- We propose an oracle-gap criterion for benchmarking communicative rewards—the gap between a baseline and an oracle given all private states bounds the task value of communication—and introduce scout-support, a tractable two-agent benchmark with a large, structural gap: a scout’s private target only pays off if a slower supporter infers it early, while a pickup waypoint makes the scout’s task-optimal motion uninformative exactly when information is needed. A cooperative-navigation control with near-zero gap completes the pair.
- Empirically, we dissect implicit communication into speaker and receiver contributions at converged training budgets. Motion-grounded hand-crafted listeners (progress, filter, plain cosine) close 60–82% of the oracle premium through the speaker-side reward alone across three structurally distinct environments with frozen hyperparameters (Welch $t \geq 4$ at 8–10 seeds), and a blind-partner control zeroes the gain while sparing the oracle—the effect is communication, not shaping, with the largest gains in the teammate’s commitment timing. A learned listener saturates (97% decoding of any behavior) and shapes nothing; crucially, we then run the ablation this observation demands—learned listeners bounded to the audience’s viewpoint and to the pre-commitment window—and find that saturation returns and the literal reward still shapes nothing at any weight, while RSA’s contrastive recursion recovers a third of the premium at a raised weight (35%, $n=16$ with pre-registered confirmation, $t = 3.4$; recursion vs. weight-matched literal control, $t \approx 3.9$; suggestive under our conservative policy, replicated in direction across four cohorts). Pragmatic contrast, not decoding, is what survives learning; what the hand-crafted winners still add is what they credit: only overt geometric signals the partner’s own perception can read. We distill this into an audience-matching principle and give it formal content as a gradient-saturation statement. We also document, as a methodological hazard, that pre-convergence evaluation of these co-adaptive systems produced repeated spurious reversals.

2. RELATED WORK

Our framework, which uses pragmatic inference to generate an intrinsic reward for communicative clarity, builds on four lines of research: (1) legibility and observer-aware planning; (2) intrinsic rewards for multi-agent coordination; (3) implicit and emergent communication; and (4) computational pragmatics.

2.1. Legibility, Goal Recognition, and Observer-Aware Planning

Inferring an agent’s latent state (e.g., its goal) from observed behavior is the central problem of Inverse Reinforcement Learning (IRL) [6] and computational Theory of Mind (ToM) [7, 8]. Legibility inverts this problem: the actor shapes its own behavior so that an observer’s inference succeeds. Originally formalized for robot motion [1], legibility has been extended to stochastic sequential decision problems [9], unified with explicability and predictability in observer-aware MDPs [10, 11], and recently brought to MARL through legible variants of interactive POMDPs [12]. Many of these approaches embed the observer model in the planning objective, requiring inference over goals at decision time; a complementary line has used observer models as reward signals inside RL—from policy search against human observers [32], to observer-in-the-loop shaping [33], legibility-regularized policies [34], and supervised observer models for legible planning [35]. The Bayesian teaching literature reaches the same mechanism from pedagogy: agents select demonstrations to communicate latent task structure to an inferring observer [30, 31]. Relative to this body of work, our contributions are the max-ent/RSA derivation of the reward, a systematic study of which listener should anchor it—the question prior work fixes by assumption—and an oracle-gap evaluation protocol that certifies the reward has task value to recover. Like this prior work, the reward is computed offline during training, so the executed policy carries its communicative competence with no observer model, belief tracking, or additional computation in the execution loop.

A complementary line of work uses ToM as a source of intrinsic motivation [13], rewarding agents for accurately predicting the beliefs of others; this is listener-centric, rewarding agents for being predictive of others. Our pragmatic framework is speaker-centric: it rewards agents for making their own private state clear, regardless of their partners’ modeling ability.

2.2. Intrinsic Rewards for Multi-Agent Coordination

Our pragmatic reward is a form of intrinsic motivation for cooperation, and is thus related to social influence

[14], which rewards an agent for causally changing its partners’ behavior. The communicative goals differ: influence rewards an agent for compelling a reaction, whereas the pragmatic reward incentivizes the successful transmission of information about oneself, decoupling the speaker’s reward from the listener’s reaction. Teammates remain free to exploit or ignore the revealed intention, which promotes a more flexible form of coordination.

The closest existing mechanism to R_{comm} is the auxiliary reward of Tian et al. [38], which pays an agent for its partner’s correct belief about private information transmitted implicitly through actions (demonstrated in bridge bidding). Our reward differs in being derived from a listener model rather than the partner’s own belief—which is what makes the “which listener” question, our central subject, visible at all: Tian et al. fix the audience to be the actual partner’s inference network, one point in the listener space we sweep, and one that our saturation analysis (Prop. 2) predicts loses shaping power exactly when the partner learns to decode. Also close is the information-regularization framework of Strouse et al. [4], which adds a mutual-information term $I(A; G|S)$ to the objective so that agents share (or hide) their goal G through their actions. Similarly, Eccles et al. [15] add a positive-signaling bias encouraging messages to depend informatively on observations, and, in the single-agent setting, DIAYN [5] rewards skills for being discriminable from states via a learned classifier. Proposition 1 makes the relationship precise: these MI-based objectives correspond to the literal listener level of our framework, obtained when the RSA recursion depth is zero and L_0 is trained by maximum likelihood. The pragmatic levels add contrastive reasoning over the alternative actions the agent could have taken—a model of communication richer than statistical dependence alone. Our framework also shares the goal of grounded, convention-free coordination with Bayesian action decoding [16] and off-belief learning [18], which control how much agents may read into their partners’ actions; we instead directly reward the actor for making the intended reading easy. The cross-play evaluations we report connect to the ad-hoc-teamwork setting [37], where partners meet without co-training.

2.3. Implicit Communication vs. Explicit Channels

A significant body of MARL research equips agents with a dedicated communication channel and learns an emergent protocol end-to-end [19, 20, 21], including channels that carry predicted intentions directly [36]. Our framework diverges from this paradigm by focusing on implicit communication [40]: there is no separate channel and no explicit message passing. An agent “speaks” by moving

in the environment, and a teammate “listens” by observing that movement. The closest formal treatment of this setting is that of Sokota et al. [29], who encode messages into trajectories via minimum-entropy coupling on top of a max-ent policy; their construction targets a chosen bit-rate through the action channel, whereas our reward asks for legibility of the task-relevant private state only insofar as it pays, and requires no coupling machinery at execution. The pragmatic reward incentivizes actions that are simultaneously effective for the task and clear as signals of intent. This is valuable when explicit communication is costly, unavailable, or must be fused with physical action, as is common in human-agent teams. Because the signal is the action, the exploration-communication conflict identified in emergent-communication work [15, 17] appears here as a reward-shaping trade-off, which we analyze through the weight λ .

2.4. Computational Pragmatics

RSA [2] formalizes communication as recursive probabilistic reasoning between a speaker and a listener, and has been given a rate-distortion interpretation [22]. Neural instantiations replace the hand-specified literal listener with learned models [41, 23, 24], a design choice our learned- L_θ instantiation transposes to the behavioral domain, and RSA-style mutual reasoning has been brought into RL-trained emergent communication over explicit message channels [39]; we instead run the recursion over actions and use its output as a reward. The Collaborative RSA (CRSA) model [3] extends RSA to multi-turn dialog between agents with asymmetric private information. We show in Section 4.6 that such purely communicative models arise as the myopic, task-free special case of our strategic framework, which extends pragmatic reasoning to long-horizon sequential decision-making where instrumental and communicative goals are intertwined.

3. FORMALISMS FOR ACTION AND INTERPRETATION

To model legible strategic behavior we require two formalisms: one for optimal sequential decision-making under uncertainty, and one for how an observer infers intent from an action. For the former we use Max-Ent RL; for the latter, RSA.

3.1. Maximum Entropy RL

Max-Ent RL extends the standard RL objective with the intuition “be as versatile as possible while still being successful.” In a single-step problem, the agent seeks a policy $\pi(a|s)$ maximizing expected reward plus policy

entropy [25]:

$$J(\pi) = \mathbb{E}_{a \sim \pi(\cdot|s)}[R(s, a)] + \alpha_{\text{RL}} H(\pi(\cdot|s)). \quad (1)$$

The optimal policy is a softmax over rewards [26], with temperature α_{RL} trading off exploitation against versatility:

$$\pi^*(a|s) \propto \exp(R(s, a)/\alpha_{\text{RL}}). \quad (2)$$

3.2. The Rational Speech Act Framework

RSA [2] models an observer who infers an agent’s intended meaning not from the literal content of an action alone, but by reasoning about why the agent chose that action over the alternatives: “if they had intended X , they would have done Y ; since they did Z , they must intend something else.” Formally, RSA seeks a communicative policy $S(u|m)$ that optimally balances communicative success against action cost, yielding a softmax over the utility, or “listener value,” of each communicative act:

$$S(u|m) \propto \exp(\alpha_{\text{RSA}} \cdot V_L(u, m)), \quad (3)$$

where $V_L(u, m)$ quantifies the clarity of act u for conveying meaning m to an observer, penalized by the act’s cost.

4. REWARDING LEGIBILITY

Our agent optimizes a composite reward balancing task objectives with communicative clarity. This section defines the reward via a formal bridge between pragmatic inference and RL, characterizes the space of listener models that anchor it, describes the architecture that computes it, and presents the resulting strategic Bellman equation.

4.1. The Total Reward

At each timestep t the agent receives the standard extrinsic task reward $R_{\text{ext}}(s_t, a_t)$ from the environment, and generates internally an intrinsic communicative reward $R_{\text{comm}}(s_t, a_t)$ quantifying how clearly action a_t communicates the agent’s private state m_t (its “meaning,” e.g., an intended target) to an imagined observer. The policy maximizes the weighted sum

$$R_{\text{total}}(s_t, a_t) = R_{\text{ext}}(s_t, a_t) + \lambda R_{\text{comm}}(s_t, a_t), \quad (4)$$

where λ balances task success against communicative clarity. The central question is how to define R_{comm} in a principled way.

Table 1. Mapping between Max-Ent RL and RSA components.

Max-Ent RL component	RSA analogue
State (s)	Context
Action (a)	Utterance (u)
Private state / intention (m)	Meaning (m)
Agent policy ($\pi(a s, m)$)	Speaker ($S(u m)$)
Intrinsic reward (R_{comm})	Listener value (V_L)
Temperature (α_{RL})	Rationality ($1/\alpha_{\text{RSA}}$)

4.2. Deriving R_{comm} via Pragmatic Inference

We treat the agent’s action a as a communicative “utterance” and its private state m as the “meaning” to be conveyed (Table 1). The optimal single-step Max-Ent policy (Eq. 2) and the RSA speaker (Eq. 3) are both softmax choice models, identical in the single-step case under the re-indexing $\alpha_{\text{RSA}} = 1/\alpha_{\text{RL}}$. We are explicit about the altitude of this observation: it is a structural correspondence, not an equivalence theorem—any Boltzmann-rational model shares the form, the identification is stated for the single-step problem while our agents are sequential, and in practice the two temperatures are decoupled (the SAC entropy coefficient is auto-tuned while the recursion of Eq. 6 is run at unit rationality). Its role is to motivate a definition: if a Max-Ent policy is an RSA speaker whenever its reward is a listener value, then legibility can be installed by making the listener value part of the reward. We define

$$R_{\text{comm}}(s_t, a_t) \triangleq V_L(a_t, m_t) = \log L_t^*(m_t|a_t, s_t) - C(a_t), \quad (5)$$

where $L_t^*(m_t|a_t, s_t)$ is the probability a pragmatic listener assigns to the agent’s true private state after observing its action in context s_t , and $C(a_t)$ is an optional action cost. In practice we set $C(a_t) = 0$: the cost of a legible action is already handled by the task reward, since a less direct but clearer path is naturally penalized by a lower R_{ext} . In our implementation we also add the constant $\log |\mathcal{M}|$, so that R_{comm} reads as the information gain over a uniform prior: it is zero when the action is uninformative and positive when the listener’s posterior concentrates on the true meaning.

4.3. The Space of Literal Listeners

Computing L^* requires anchoring the pragmatic recursion in a literal listener $L_0(m|a, s)$: the assumed common ground specifying how behavior would be interpreted absent pragmatic reasoning. Given L_0 and a set $\mathcal{A}$ of alternative actions (the true action plus K sampled alternatives; we sample uniformly from the action space unless noted, and study policy-sampled alternatives in

Sec. 6.1.1), the pragmatic listener is obtained by N_r steps of RSA alternation at unit rationality and uniform meaning prior,

$$S_k(a|m) = \frac{L_{k-1}(m|a, s)}{\sum_{a' \in \mathcal{A}} L_{k-1}(m|a', s)}, \quad L_k(m|a) = \frac{S_k(a|m)}{\sum_{m'} S_k(a|m')}, \quad (6)$$

and $L^* = L_{N_r}$. Because the alternative set is sampled, R_{comm} is a stochastic reward whose sampling noise the critic averages over. When alternatives are drawn from the current policy—or when L_0 itself co-adapts, as with the learned listener below— R_{comm} is additionally policy-dependent and hence non-stationary; the Bellman analysis of Sec. 4.5 then holds per reward snapshot, as in other co-adaptive intrinsic rewards [5]. The choice of L_0 is the framework’s most consequential free parameter, and we study four instantiations.

(i) Direction listeners. A computationally cheap heuristic interprets actions by their geometric similarity to ideal paths. For agent position p and each meaning m , let v_m be the unit vector toward the nearest target consistent with m . A naïve score is $S(a, m) = \alpha_{\text{temp}} \cos(a, v_m)$ (simple). To capture unambiguity explicitly, an exclusivity variant penalizes similarity to competing meanings,

$$S_{\text{excl}}(a, m) = \alpha_{\text{temp}} (\cos(a, v_m) - \max_{m' \neq m} \cos(a, v_{m'})), \quad (7)$$

and $L_0(\cdot|a) = \text{softmax}_m S_{\text{excl}}(a, \cdot)$. This listener rewards actions aligned with the ideal path toward the true target and misaligned with the ideal paths of all confusable targets.

(ii) Progress listener. Instead of instantaneous direction, this listener scores the progress an action realizes toward each candidate target: $S_{\text{prog}}(a, m) \propto d_{t-1}(m) - d_t(m)$, the per-step reduction in distance to the target consistent with m , normalized by the maximum step length. This mirrors the cost-to-go formulation of legibility in robot motion [1], which scores partial trajectories by how efficiently they advance each candidate goal.

(iii) Filter listener. Instantaneous listeners ignore history: an action that is uninformative in isolation may complete a legible trajectory. The filter listener maintains a recursive belief over meanings, updated multiplicatively from per-step directional likelihoods with exponential forgetting,

$$b_t(m) \propto b_{t-1}(m)^\rho \exp(\alpha_{\text{temp}} \cos(a_t, v_m)), \quad (8)$$

and rewards the log-belief in the true meaning, $R_{\text{comm}} = \log b_t(m_t) + \log |\mathcal{M}|$. Under discounting this reproduces the trajectory-integral character of legibility—early clarity is worth more than late clarity—and makes R_{comm} explicitly non-Markovian, which the PRM accommodates because rewards are computed offline.

(iv) Learned listener. Hand-crafted listeners presuppose that the features they score are the ones an actual observer would use. To remove this assumption, we train a network $L_\theta(m|s, a)$ by maximum likelihood on the replay buffer to predict an agent’s true meaning from its publicly observable context and action (the private state is masked from the input). The listener is trained concurrently with the policy, so the common ground co-adapts: the reward only credits signals that are actually decodable from the agents’ current behavioral distribution. With $N_r = 0$ the reward is simply $R_{\text{comm}} = \log L_\theta(m_t|s_t, a_t) + \log |\mathcal{M}|$; with $N_r \geq 1$ the RSA recursion (Eq. 6) is applied on top of L_θ , adding contrastive reasoning over alternatives.

The learned instantiation admits an information-theoretic characterization.

Proposition 1. Let meanings be drawn uniformly on $\mathcal{M}$, and let $L_\theta(m|s, a)$ be any conditional distribution. Then, under the joint distribution induced by the policy,

$$\mathbb{E}[\log L_\theta(m|s_t, a_t) + \log |\mathcal{M}|] \leq I(M; S_t, A_t), \quad (9)$$

with equality iff $L_\theta(m|s, a) = p(m|s_t = s, a_t = a)$. Hence the expected learned-listener reward is the Barber–Agakov variational lower bound [27] on the mutual information between the agent’s private meaning and its observable behavior; maximum-likelihood training of θ tightens the bound, and policy optimization against it maximizes information disclosure.

Proof. $\mathbb{E}[\log L_\theta(m|s, a)] = \mathbb{E}[\log p(m|s, a)] - \mathbb{E}_{s,a}[\text{KL}(p(\cdot|s, a) \| L_\theta(\cdot|s, a))]$. Adding $H(M) = \log |\mathcal{M}|$ gives the claim; the KL term vanishes iff L_θ matches the true posterior. $\square$

Three remarks locate Proposition 1 precisely. First, it holds for any conditional distribution over the same conditioning variables: the instantaneous hand-crafted listeners of (i)–(ii) also yield variational lower bounds on the same mutual information, only looser ones (the filter listener, which conditions on the action history, instead bounds the larger quantity $I(M; S_{1:t}, A_{1:t})$ and can exceed the instantaneous information). What distinguishes the learned listener is not being a bound but being the tightest bound available from the data; the empirical calibration results of Sec. 6.3.2 are exactly bound-tightness. Second, the connection to prior objectives is close but not exact: the skill discriminator of DIAYN [5] maximizes a bound of this form, while the cooperative information regularizer of Strouse et al. [4] targets the conditional information $I(A; G|S)$ —which differs from $I(M; S, A)$ by the state-borne term $I(M; S)$ —and the positive-signaling bias of Eccles et al. [15] shapes an explicit message channel with entropy targets rather than a variational decoder. Within our framework all three

are forms of the literal ($N_r = 0$) level of pragmatic reasoning with a learned common ground. Third, the RSA recursion scores an action not by how much information it carries in isolation but by how it contrasts with the alternatives the agent declined [2, 22]; since the recursion moves L^* away from the maximum-likelihood fit, it loosens the expected-reward bound of Proposition 1 whenever that fit is at the posterior (with finite data the direction is not guaranteed, but tightening is not its purpose)—its value, where we find any, is as a shaping signal, not as an information estimate (Sec. 6.1.1).

The proposition also delivers the failure mode that dominates our experiments. If the listener already decodes the agent’s meaning on all behavior the policy actually produces, the reward is a constant on the policy’s support and cannot shape it:

Proposition 2 (Gradient saturation). Fix a listener L and a policy π_ϕ with occupancy measure ρ_π , and suppose the reward is clipped below at $-B$ (i.e., $R_{\text{comm}} = \max(\log L(m_t|s, a), -B - \log |\mathcal{M}|) + \log |\mathcal{M}|$). If $\rho_\pi(L(m_t|s, a) \geq 1 - \epsilon) \geq 1 - \delta$, then for any score-function (policy-gradient) estimator with baseline $b = \mathbb{E}[R_{\text{comm}}]$,

$$\|\nabla_\phi \mathbb{E}_{\pi_\phi}[R_{\text{comm}}]\| \leq (-\log(1-\epsilon) + \delta(B + \log |\mathcal{M}|)) \mathbb{E}_{\pi_\phi} \|\nabla_\phi \log \pi_\phi\|, \quad (10)$$

which is $O(\epsilon + \delta B)$ per step; over a discounted horizon the constant is scaled by at most $1/(1 - \gamma)$.

Proof. On the $(1 - \delta)$ -mass, $R_{\text{comm}} \in [\log(1 - \epsilon) + \log |\mathcal{M}|, \log |\mathcal{M}|]$, an interval of width $-\log(1 - \epsilon)$; on the remaining δ -mass the clipped reward deviates from b by at most $B + \log |\mathcal{M}|$. The estimator $\mathbb{E}[\nabla \log \pi \cdot (R_{\text{comm}} - b)]$ is bounded by the expected deviation of the reward from b ; $-\log(1 - \epsilon) = \epsilon + O(\epsilon^2)$. $\square$

Two scope notes. The high-probability form is the one our experiments instantiate: a Gaussian policy has unbounded support, so a pointwise bound would be vacuous, but the measured decode accuracies (Table 2) are exactly estimates of $1 - \delta$ at small ϵ . And our agents are trained with reparameterized (SAC) gradients rather than score-function estimators; there the same interval argument bounds the spread of the reward entering the critic targets, so a saturated listener shifts Q by a near-constant that policy improvement cancels. We therefore use Proposition 2 as the mechanism consistent with the observed inertness rather than a quantitative prediction. It formalizes the saturation direction of the audience-matching principle developed in Sec. 7—a listener that decodes everything demands nothing—while the principle’s second clause (crediting the right vocabulary) remains empirical. Note the antecedent is a property of the listener under the policy’s occupancy—a listener can saturate either by being given context that reveals the meaning regardless of the action (as our full-context

learned listener does) or by co-adapting fast enough to decode whatever the current policy emits (as our viewpoint-bounded one does, Sec. 6.1.1).

4.4. The Pragmatic Reward Module

The theoretical bridge yields a practical hybrid architecture that decouples action selection from communicative analysis. The system comprises the main RL agent, which learns $\pi(a_t|s_t)$, and a Pragmatic Reward Module (PRM), which computes R_{comm} to shape the agent’s learning updates. The pragmatic analysis is performed offline during training—for static listeners, when a transition is collected; for the learned listener, when it is replayed, so the reward tracks the co-adapting common ground—and never in the action-selection loop, adding no latency to decision-making. We operate under CTDE: during centralized training the PRM has access to all agents’ private states, enabling a gold-standard legibility reward; at execution, agents act from local observations only, and the PRM is discarded.

4.5. The Strategic Bellman Equation

With the composite reward defined, the action-value function of the legible agent satisfies the soft Bellman equation

$$Q_t(s_t, a_t) = R_{\text{total}}(s_t, a_t) + \gamma \mathbb{E}_{s_{t+1} \sim P(\cdot|s_t, a_t)} \left[\alpha_{\text{RL}} \log \int \exp \left(\frac{1}{\alpha_{\text{RL}}} Q(s_{t+1}, a') \right) ds_{t+1} \right] \quad (11)$$

which is the standard Max-Ent backup applied to R_{total} . The equation makes explicit the trade-off between immediate gains (task and clarity) and the long-term strategic value of future states: an agent may accept a lower immediate reward (e.g., a “corralling” detour) if the resulting state carries higher expected value through improved team coordination. Two caveats scope this equation. For co-adapting listeners the reward is non-stationary, so Eq. 11 describes the fixed-point problem at each reward snapshot rather than a single stationary MDP—the standard situation for learned intrinsic rewards [5]. And the filter listener’s reward is a function of the belief state, hence non-Markovian in (s_t, a_t) ; a Markovian critic trained against it absorbs the history-dependence as reward noise. We flag both as gaps between the clean theory and the working system.

4.6. Generality: Recovering Communicative Agents

Purely communicative agents arise as a special operating point. Setting $R_{\text{ext}} = 0$ and $\lambda = 1$ makes the total reward purely communicative; setting $\gamma = 0$ makes the

agent myopic, collapsing Eq. 11 to the immediate listener value plus policy entropy—a soft-max speaker over V_L , which is the same objective form optimized per turn by dialog models such as Collaborative RSA [3]. We state this as a correspondence of objective forms rather than a formal reduction: a full mapping would additionally have to align CRSA’s history-conditioned listener and utterance alternatives with our one-step V_L , which we do not undertake here. The point of the observation stands at the level of the framework: purely communicative pragmatic agents occupy the myopic, task-free corner of the same composite objective that our strategic agents optimize over long horizons.

5. EXPERIMENTAL SETUP

5.1. Environments

A meaningful benchmark for communicative rewards must certify that communication has task value at all. We therefore adopt an oracle-gap criterion: comparing a baseline against an oracle that receives all private states in its observations measures the maximum task reward that intent transmission can unlock. We evaluate on two environments, one on each side of this criterion.

Environment A: scout-support (converged oracle gap 0.112 per step; Sec. 6.1). A two-agent team in a continuous 2D arena, modeled on scout-and-support patterns in search-and-rescue and warehouse logistics: a fast scout privately knows which of three sites is the true target, and a slower supporter earns the team a bonus only while both agents occupy the target together. Crucially, the scout must first visit a central pickup waypoint before its presence at the target counts (a forklift fetches the pallet before the dock). During this first leg, the scout’s task-optimal motion is identical regardless of target—structurally uninformative—so any early information the supporter receives must be actively signaled, e.g. by curving the approach to the waypoint toward the future destination, the anticipatory motion studied in robot legibility [1]. Sites lie on a circle (radius 1.0, 120° apart, jittered); the supporter’s lower speed (0.6 vs. 1.0) makes late commitment costly. The reward is the rendezvous bonus minus both agents’ distances to their current objectives; episodes last 50 steps. The supporter observes the scout’s position and velocity but never its target. Meanings are the three sites; only the scout receives R_{comm} . Two structural variants test generality—a five-meaning version (sites at 72°; harder inference) and a minefield version (entering a non-target site is penalized; wrong commitment is costly rather than slow)—and a blind control masks the supporter’s view of the scout to test whether gains flow through the channel at all.

Environment B: preference spread (oracle gap ≈ 0). A cooperative-navigation control: three agents and three colored landmarks, each agent holding a private color preference; the team is rewarded for matched coverage, penalized for distance and collisions. Because proximity resolves most coordination and acting on one’s own preference is extrinsically rewarded, an oracle gains nothing—communication here has no task value to unlock, which is precisely what makes it a useful control for measuring what a communicative reward costs.

5.2. Models and Conditions

All agents are trained with multi-agent Soft Actor-Critic (MASAC)—parameter-shared Gaussian actors (roles distinguished by a one-hot flag) and a centralized twin critic over joint observations and actions [28], with automatic entropy tuning—the natural multi-agent extension of the Max-Ent framework our theory is built on. The critic is trained on the scalar team reward with λR_{comm} added; R_{comm} is computed from the scout’s action and paid into that team scalar, so “only the scout receives R_{comm} ” means the reward is a function of the scout’s behavior, credited through the shared critic. Conditions differ only in reward and observation content: baseline ($\lambda = 0$); oracle ($\lambda = 0$, private states shared through observations); the four hand-crafted listeners of Section 4.3—simple, exclusivity (both with one RSA step over $K = 64$ sampled alternatives), progress, and filter; and the two learned ones, learned ($N_r = 0$) and learned + RSA (one recursion step over 16 alternatives). L_θ is a 2-layer MLP trained by cross-entropy on replayed transitions with the private state masked from its input. In Environment A we add ear conditions that append the listener’s posterior to the supporter’s observation at execution (Section 6.1), bounded learned conditions that restrict the learned listener’s inputs and training window to what the audience actually uses (Section 6.1.1), and a value-of-information weighting that scales R_{comm} by the supporter’s decision relevance (full weight before the scout’s pickup, down-weighted after). Environment B uses $\lambda = 0.1$ over 150 training cycles (480k steps) with the direction-based and learned listeners; Environment A uses $\lambda = 0.3$ over 400 cycles, with the headline conditions run at 10 seeds and the rest at 3–8 as noted per table. We disclose hyperparameter provenance: $\lambda = 0.3$ and the VOI weight 0.2 were selected in preliminary short-budget sweeps on the standard scout-support task itself, then frozen; the transfers to the five-meaning, minefield, and boundary-sweep settings are therefore untuned, while the standard-task numbers are reported on the environment where the two knobs were chosen. Full hyperparameters are in Table 6, and the launch specifications in the released code determine every run

exactly.

5.3. Evaluation Metrics

We evaluate periodically on a fixed set of 64 evaluation episodes (identical layouts across all conditions and evaluations, making comparisons paired) with deterministic policies; final performance averages the last three evaluations. We report per-condition means $\pm$ s.e.m. over seeds with Welch t -tests, and adopt the following multiple-comparison policy: claims we label as headline results have $t \geq 4$ (surviving a Bonferroni correction across all ~ 30 tests in the paper at $\alpha = .05$), while effects at conventional $p < .05$ are reported as suggestive and marked as uncorrected. Task performance is the mean per-step team reward and its components. In Environment A, commitment accuracy is the fraction of steps in which the supporter’s nearest site is the true target—a direct behavioral readout of how early it decodes the scout. In Environment B, team specialization is the fraction of landmarks claimed uniquely (each agent assigned to its nearest landmark); a score of 1 indicates perfect division of labor. Legibility is measured by a fixed probe listener—the simple geometric listener, used identically across all conditions—reporting the accuracy with which an agent’s true preference is recovered from its instantaneous action (probe intent accuracy). The probe is computed over all steps, including those where an agent already rests on its target and its actions are uninformative, so absolute values understate legibility; because the probe is fixed and external to all training conditions, however, relative improvements reflect genuinely clearer behavior rather than adaptation to the metric—with one disclosed exception: for the simple-listener condition the probe coincides with its own training reward, so for that condition we rely on the post-hoc decoder below instead. For learned-listener conditions we additionally report L_θ ’s own held-out accuracy, and for all PRM conditions the achieved communicative reward R_{comm} . Finally, we measure post-hoc decodability: for every trained run, a fresh listener network is trained from scratch on deterministic rollouts of the final policy and evaluated on held-out episodes. This quantifies how much intent information each condition’s converged behavior carries under an identical decoding budget, independently of any listener used during training.

6. RESULTS

We present Environment A—where the oracle gap certifies that communication has task value—as the primary benchmark, and Environment B as the control that isolates what a communicative reward costs when there is nothing to say. Across both, the results separate cleanly

along the common-ground axis: what matters is not whether a pragmatic reward is used, but which listener anchors it.

6.1. Environment A: When Communication Pays

Table 4 summarizes Environment A at the converged budget (400 cycles; 10 seeds for all key conditions); Figure 4 shows the time course of the supporter’s understanding and Figure 5 the information carried by the scout’s behavior. The converged oracle premium is 0.112 per step (0.957 vs. 0.845; commitment .880 vs. .851)—the irreducible value of early intent information once both teams have fully learned the task.

Motion-grounded hand-crafted listeners close most of the recoverable gap. The strongest conditions are the simplest: the progress listener closes 70% of the oracle premium ($0.924 \pm .008$, Welch $t = 6.2$ vs. baseline), the filter listener 67% ($t = 6.5$), and even the plain cosine listener 62% ($t = 5.0$)—through the speaker-side reward alone, with the largest commitment gains of any condition (.857–.861) and visibly more legible scouts (post-hoc decoder accuracy .61 vs. .54; time-resolved in Fig. 5). The mechanism is direct: these listeners only credit overtly geometric signals—motion biased toward the true target—and overt signals are exactly what the supporter’s own policy network can read from raw observations and motion history. No dedicated receiver machinery is needed; the channel is the behavior itself. The exception among the hand-crafted family proves the calibration rule: the exclusivity listener, whose margin demand is geometrically unsatisfiable here as in Environment B, is unstable across seeds ($0.795 \pm .081$ at $n=8$; individual seeds range from baseline-level to collapsed)—an early $n=3$ estimate had suggested a positive 16% closure, which the extension eliminated.

A blind-partner control certifies the mechanism. Intrinsic rewards densify learning signals, so any gain could in principle be reward shaping rather than communication. We sever the channel: a blind supporter whose observation masks the scout’s position, velocity, and motion history (8 seeds; blind oracle 4). The pragmatic reward’s gain does not merely vanish—it inverts into a significant cost: blind progress-listener teams score below blind baselines (-1.67 vs. -1.56 ; Welch $t = 2.8$, $p \approx .02$, uncorrected), because the scout still pays the signaling detour for messages nobody receives. Meanwhile the blind oracle, which needs no channel, retains full performance (0.963 vs. 0.957 sighted). Every unit of the recipe’s improvement flows through the partner’s observation of the speaker’s motion: the effect is communication, not shaping—and the residual, once the channel is cut, is pure cost.

A too-competent listener exerts no pressure. The

learned listener inverts this logic and fails by succeeding. Trained to convergence, L_θ decodes the scout at 97% whatever the scout does: its reward saturates near the log 3 ceiling (+1.03), the pragmatic gradient on behavior vanishes, and the speaker-only learned condition ends indistinguishable from—nominally below—the baseline (0.828, -15% closure, n.s.). Deploying the same listener as the supporter’s ear (its posterior appended to the supporter’s observation, still decentralized-executable) recovers 30% of the premium ($0.878 \pm .012$, Welch $t = 2.15$, $p \approx .05$ uncorrected—suggestive under our multiple-comparison policy), and the ear works nearly as well with no speaker subsidy at all (0.874, 26%): with a saturated listener, all the value is in the decoding, none in the speaking. Combining the two families—the filter reward for the scout with its belief as the ear—matches the filter reward alone (0.920, 66%): once the speaker signals overtly, the equipped ear is redundant, confirming that the overt channel is read directly from behavior.

6.1.1. Bounding the Learned Listener: The Ablation the Principle Demands

If the learned listener fails because it is too strong, the audience-matching principle predicts a specific remedy: weaken it toward the intended reader. We test this directly, bounding the learned listener along the two axes RSA itself suggests (Table 2). A viewpoint-bounded listener sees only what the audience uses to interpret the utterance—the site bearings and the scout’s action—rather than the full context and motion history; this removes the contextual shortcut (its accuracy now starts at chance rather than 84%). A window-bounded variant is additionally trained only on pre-commitment steps, so decode accuracy can no longer be earned from the post-commitment leg, where the scout’s motion trivially reveals the target.

The first result is a negative control that sharpens the principle. Both bounded listeners re-saturate: without the contextual shortcut, saturation merely arrives later (decode accuracy 0.54 at a quarter of training for the window-bounded listener, versus 0.86 viewpoint-only and 0.88 full-context) and by a different route—the listener co-adapts to whatever the current policy emits, exactly the second saturation mode anticipated by Proposition 2. The literal bounded rewards end at baseline parity, curing the full-context listener’s harm but recovering none of the premium. Layering the RSA recursion on top changes the picture in a small but consistent way: in two replications (same eight seeds, independently trained under the two bounding schemes) the pragmatic variant lands at 0.868 (a 21% closure each time), ahead of its literal counterpart at all but one evaluation point

Table 2. Bounding the learned listener (Environment A, 400 cycles, 8 seeds per condition except the first and last rows, which repeat Table 4’s $n=10$ entries; premium = 0.112; recipe $\lambda = 0.3$ unless noted). “Sat.” is the listener’s converged decode accuracy over the episode; for +RSA rows, R_{comm} metrics elsewhere report the literal listener value. The decisive contrast is the $\lambda=0.6$ block: recursion versus no recursion at the same weight is worth 0.064 ($t \approx 3.9$; the $n=16$ row pools the eight selection seeds with a pre-registered eight-seed confirmation, reported separately in the text).

Condition	R_{ext}	Closure	Commit	Sat.
Learned (full context)	$0.828 \pm .015$	−15%	.853	.97
+ viewpoint bound	$0.849 \pm .017$	4%	.851	.99
+ viewpoint, + RSA	$0.868 \pm .011$	21%	.856	.95
+ window bound	$0.841 \pm .017$	−4%	.853	.94
+ window, + RSA	$0.868 \pm .009$	21%	.856	.95
Window bound, $\lambda=0.6$	$0.820 \pm .016$	−22%	.854	.94
+ RSA, $\lambda=0.6$ ($n=16$)	$0.884 \pm .006$	35%	.861	.94
+ RSA, $\lambda=1.0$	$0.849 \pm .013$	4%	.864	.96
Progress (hand-crafted)	$0.924 \pm .008$	70%	.861	.96

across training.

The second result turns the consistent trend into the paper’s cleanest causal contrast. Raising the weight— $\lambda = 0.6$, the middle of three values swept—lifted the recursed variant to $0.893 \pm .009$ (43% closure) on the selection runs; because that estimate rides a swept knob, we then ran a pre-registered confirmation on eight fresh seeds at the fixed $\lambda = 0.6$ and observed the regression to the mean our own Sec. 6.2.1 teaches readers to expect: $0.875 \pm .005$ (27%, $t = 2.7$). The honest estimate pools both cohorts: $0.884 \pm .006$, a 35% closure at $t = 3.4$ against baseline, with commitment accuracy .861 equal to the progress listener’s. (We initially motivated the higher weight by the pragmatic reward’s $\log |\mathcal{M}|$ ceiling; since all listener values share that ceiling, the honest statement is that the effective scale of the gradient-carrying variation differs across listeners, and λ must be set against it empirically.) The control is decisive: the literal window-bounded listener at the same $\lambda = 0.6$ falls below baseline ($0.820 \pm .016$), consistent with the saturation mechanism of Proposition 2—amplifying a saturated reward amplifies noise and signaling cost, not signal. Recursion versus no recursion at matched weight is worth 0.064 ($t \approx 3.9$ at pooled $n=16$ vs. 8). Under our conservative multiple-comparison policy (Sec. 5.3) these fall just short of headline grade and we label them accordingly—suggestive, but replicated in direction across four independent cohorts (two bounding schemes, the lever sweep, and the confirmation). The

dose-response is inverted-U ($\lambda = 1.0$ falls back to 0.849, though with the highest commitment accuracy of any learned condition, .864), and two further levers are inert: drawing RSA alternatives from the current policy instead of uniformly matches the uniform variant ($0.872 \pm .013$), and slowing the listener’s adaptation ($10\times$ smaller learning rate) neither helps nor hurts (0.861). The alternatives result nuances the mechanism story: if recursion’s contrast came only from off-distribution alternatives, policy-sampled ones should approach the inert literal condition, and they do not—at 94% decode accuracy the listener is saturated enough to silence the literal alternatives to leave the contrastive term informative. A quantitative account of this margin is open.

Why the residual gap to the hand-crafted family (35% vs. 70%)? The commitment ladder locates it: baseline .851, bounded-learned+RSA .861, progress .861, oracle .880—on commitment the recursed listener has caught up, but its task reward still trails, and the learned listener remains satisfied by any decodable regularity, including idiosyncratic codes that the actual audience—a co-training feed-forward supporter—tracks imperfectly; the progress listener credits only monotone geometric approach, which anything watching raw motion can read. The constructive summary for learned reward listeners is a recipe: bound the inputs to the audience’s viewpoint, train on the decision window, add the pragmatic recursion, and calibrate the weight against the reward’s gradient-carrying variation. Each ingredient is prescribed by RSA’s own structure, and ablating any one of them costs most of the gain. Finally, we report a corrected condition: deploying the viewpoint-bounded listener’s posterior as the supporter’s ear initially appeared catastrophic (0.551) due to a train/eval asymmetry in our harness (evaluation omitted the posterior injection the supporter was trained with—a bug we disclose because it shipped in an earlier code snapshot); re-evaluating the trained policies with the injection restored gives baseline parity ($0.841 \pm .023$, n.s.). The bounded ear neither helps nor hurts, which is itself informative: unlike the full-context listener, whose posterior gives the supporter information it lacks (the suggestive 30% ear gain above), the viewpoint-bounded listener sees only what the supporter already sees, so its posterior is redundant.

One recipe, three environments. Figure 3 summarizes the paper’s central comparisons on a single axis: the fraction of each environment’s oracle premium closed. To rule out benchmark idiosyncrasy, we transfer the winning recipe—progress listener, $\lambda = 0.3$, VOI weighting, every hyperparameter frozen—to two structurally distinct variants, each re-gated for its own oracle premium (8 seeds throughout). With five meanings (sites

at 72°, chance = 0.2), inference is harder and the recipe closes 82% of the premium (0.787 vs. baseline 0.725 and oracle 0.801; Welch $t = 4.0$, $p = .001$). In a minefield variant where entering a wrong site is penalized—changing the risk structure so hedging is costly rather than merely slow—it closes 61% (0.898 vs. 0.819/0.947; $t = 5.2$, $p < .001$). Together with the 70% closure in the standard task, the recipe delivers majority-gap closure at $p \leq .001$ under varied meaning-space size and payoff structure, with no per-environment tuning.

Robustness and grounding. At the converged budget the progress recipe is insensitive to its two knobs across the moderate range: 0.921–0.925 for $\lambda \in \{0.1, 0.3\}$ with VOI weighting on either setting that includes it (0.895 at $\lambda=0.1$ with uniform weighting; 3 seeds per new cell), so the flagship 70% closure is a plateau, not a tuned point. The plateau has a cliff: $\lambda = 0.6$ is destructive for the progress reward ($0.450 \pm .203$; individual seeds collapse), while the same λ is the optimum for the bounded learned-pragmatic reward (Sec. 6.1.1)—the safe weight range scales inversely with the reward’s magnitude, which is itself an argument for bounded rewards. A parallel surface measured on the learned-ear condition is flat (0.83–0.88, $\lambda \in [0.1, 0.6]$) until $\lambda = 1$ (0.61); the weighting effects that mattered mid-training (Sec. 6.2.1) wash out asymptotically. Zero-shot cross-play—pairing scouts and supporters from independently trained seeds—retains 92–94% of self-play reward in every condition, no worse than the baseline’s own cross-seed transfer; cross-paired filter teams (0.844) still match baseline self-play. The emergent protocols are thus grounded in task geometry rather than seed-specific convention, addressing the fragility concern raised by the zero-shot-coordination literature [18]. Finally, at $16\times$ the parameter count (hidden width 1024, 3 seeds) the qualitative ordering is unchanged—learned $\approx$ baseline $<$ ear $<$ oracle—though larger networks shrink the premium itself (0.945 vs. 1.008) by coordinating better from raw observations.

6.2. When Does Pragmatism Pay? A Boundary Map

The results so far establish that the recipe works in three environments and fails in a fourth (Environment B). To turn that into a predictive statement rather than a list, we vary the two quantities that theory says should govern it—how much early intent is worth and how much signaling costs—and ask how the pragmatic gain responds. We sweep the supporter’s speed (0.45–0.90, which sets how badly late commitment hurts and hence the oracle premium) and the scout’s distance-shaping weight ($2\times$, $4\times$, which sets the reward price of a signaling detour), 5 seeds per condition, everything else frozen. Plotting every setting we ran on a single pair of axes—premium

against gain—collapses the whole study into one picture (Fig. 1).

An empirical regularity, stated with its scope. Wherever the premium is statistically resolvable (point estimate exceeding three times its standard error, fixed before the analysis), the recipe captures a roughly constant fraction of it. An origin-constrained fit over all seven resolvable settings—spanning meaning-space size (3, 5), payoff structure (minefield), and partner speed—gives

$$\text{gain} \approx 0.51 \times \text{premium} \quad (\text{bootstrap 95\% CI } [0.42, 0.59]), \quad (12)$$

but this pooled constant should be read with care, for two reasons we make explicit. First, the two settings we separately classify as failure modes below (ratio 0.37) have the largest premiums and therefore dominate an origin-constrained fit; excluding them, the five remaining settings give $\text{gain} \approx 0.68 \times \text{premium}$ (CI [0.57, 0.79]), with individual ratios of 0.59–0.82. We are explicit that this second fit is exploratory: the failure-mode classification was made after observing which settings fall below the line (partner speed is a continuous axis with no pre-specified cutoff), so the 0.68 carries selection bias its CI does not reflect; 0.51 over all resolvable settings is the primary estimate. The honest summary is two-tier: the recipe converts roughly two-thirds of the premium where its preconditions hold, degrading to roughly a third where the partner is too slow or signaling too costly. Second, gain and premium share each setting’s baseline estimate, so their errors are positively correlated by construction; the bootstrap CIs resample seeds within settings and account for this. All nine settings come from a single environment family and one algorithm, so we present this as an empirical regularity of that family, not a law; it would be falsified by an environment with a resolvable premium and healthy channel whose capture ratio falls outside the fitted range. What the regularity claims is modest but useful: pragmatic rewards do not add a fixed bonus—they convert a roughly predictable fraction of whatever early information is worth, without per-environment tuning.

Nothing to win, nothing won. The regularity’s corollary is its own failure condition. Where no premium is resolvable, no gain appears: in the preference-spread control the oracle is not reliably better than the baseline and the pragmatic gain is $+0.02 \pm .05$ ($t = 0.5$); at $4\times$ signal cost the task collapses into a rush to the waypoint, the premium becomes unmeasurable ($0.015 \pm .166$) and the gain is not significant ($t = 1.4$). Pragmatic rewards are not a general-purpose performance additive; they are a lever on a specific quantity, and when that quantity is zero the lever moves nothing. This is why we advocate reporting the oracle premium as a gating statistic.

Two ways to have a premium you cannot capture. Two settings fall conspicuously below the well-behaved

Table 3. When pragmatism pays. Each row is a condition we varied to the point of failure; “gain” is relative to the matched baseline.

Requirement	Violated by
Premium exists	Pref. spread; $4\times$ cost
Partner observes speaker	Blind partner
Partner can act in time	Partner speed 0.45
Signal is affordable	$2\times$ shaping
Listener credits readable code	Literal learned L_θ , any λ
Listener is calibrated	Exclusivity L_0 (unstable)
Weight matches reward scale	$\lambda=1.0$ (learned); $\lambda\geq 0.6$ (prog.)
Evaluated at convergence	150 cycles
All hold	3 environments

settings’ line, both at ratio 0.37 despite having the largest premiums we measured. When the partner is very slow (0.45), early information is worth the most (0.155) but the recipe captures least: the scout cannot signal before it has moved, and by the time the signal exists a slow partner can no longer exploit it—the premium is real but partly unreachable at any signaling latency. When signaling is expensive ($2\times$ shaping), the scout rationally emits less of it, and capture falls correspondingly. We note these two settings also have the lowest absolute baselines, so we cannot fully separate “slow partner / costly signal” from “harder task” with these runs; the five-meaning environment argues against the simple difficulty account (baseline 0.725, yet the highest capture at 0.82), but a cleaner factorization is future work.

No channel, pure cost. Finally, the blind control marks the qualitative edge of the map: with the supporter’s view of the scout masked, the pragmatic reward does not merely stop helping, it hurts (-0.11 , Fig. 1 right), because the speaker still pays for signals nobody receives. The gain is exactly as large as the channel it flows through.

Table 3 consolidates the map, including the conditions established earlier in this section. Read as a checklist, it says that a pragmatic reward pays exactly when six things hold at once: the partner cannot already infer the intent (premium), it can observe the speaker (channel), it can still act on what it learns (latency), signaling is affordable, the listener anchoring the reward is bounded, and the weight is moderate—with the whole thing evaluated at convergence. Each row is a way we made the method fail.

6.2.1. Training Dynamics: Three Stories Before the True One

We report, as a methodological caution, that this benchmark produced three successive spurious conclusions

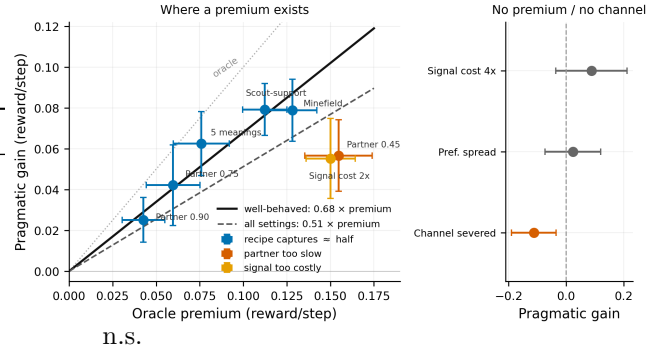


Fig. 1. The boundary map. Left: across every setting with a resolvable oracle premium—three environments, three partner speeds, two signal costs—the pragmatic gain scales with the premium (solid fit: well-behaved settings, $0.68\times$; dashed: all settings pooled, $0.51\times$; dotted line is the oracle, i.e. capturing all of it). Settings where the partner is too slow to act on a signal, or where signaling is too expensive to emit, fall below the line despite large premiums. Right: where no premium is resolvable the gain is not significant, and where the channel is severed the reward is pure cost. Left bars are ± 1 s.e.m., right bars are 95% CIs, over 5–10 seeds.

before the converged one, each supported by earnest-looking evidence. Figure 2 makes the hazard visible: the same experiment, evaluated at increasing training budgets, estimates the progress listener’s effect as strongly negative through the budget our early experiments used ($\approx -50\%$ closure at 0.5M steps), and $+70\%$ only past $\sim 1\text{M}$ steps. At 150 training cycles (of the ~ 400 needed), hand-crafted listeners appeared destructive (-1.6 for exclusivity) and the learned-ear conditions appeared to close 30% of a then-larger gap—an effect that evaporated (-3%) when seeds were doubled, because every condition was still climbing and seed-to-seed differences in learning stage masqueraded as treatment effects. Mid-training, value-of-information weighting of R_{comm} (paying for clarity before the scout’s pickup, when the supporter’s decision hangs) measurably steered emergence where uniform weighting bought only lazy clarity—decodability where behavior was already informative; at convergence both effects vanish. Communicative-MARL comparisons are unusually exposed to this hazard because the quantity of interest is a co-adaptation between policies and listeners, which converges more slowly than either alone. We recommend reporting the training-stage sensitivity of any claimed communication effect, in the form of Figure 2.

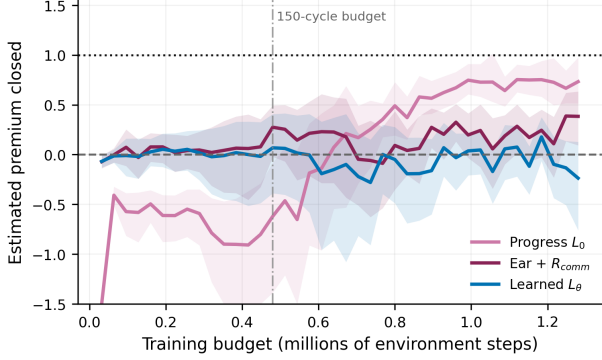


Fig. 2. The pre-convergence hazard: the estimated fraction of the premium closed, as a function of the training budget at which the comparison is made (bootstrap 95% bands, 10 seeds). The progress listener would be judged strongly harmful at our early evaluation budget (vertical line) and for the first ~ 0.7 M steps; the ordering among conditions stabilizes only past ~ 1 M steps.

6.3. Environment B: The Cost of Signaling with Nothing to Say

Table 5 summarizes final performance in the control environment; Figures 9 and 10 show the two training dynamics that carry information here (task reward, where the exclusivity listener’s collapse is visible, and each condition’s own communicative reward, where calibration levels separate).

6.3.1. Hand-Crafted Listeners: Miscalibrated Signaling

The exclusivity listener exhibits an instructive failure mode. The composite reward is clearly operative—early in training it even accelerates learning (Fig. 9), and it drives the geometric probe accuracy up faster than any other condition—yet the agent never comes close to optimizing the communicative reward itself. R_{comm} plateaus around -2.4 (Fig. 10), below the uninformative level of zero: the pragmatic listener built on this common ground assigns the true meaning less probability than a uniform guess. The exclusivity score (Eq. 7) demands that an action align with the true target’s direction while misaligning with every competing direction; when landmarks subtend similar bearings this margin is geometrically unsatisfiable, so the agent pays a permanent tax for a signal that cannot be produced. The tax is visible everywhere: final task reward drops by 46% relative to the baseline, distance and collision penalties rise, and specialization saturates lower (.815 vs. .847). The simple cosine listener, which removes the exclusivity margin, is correspondingly less harmful ($R_{\text{ext}} = 0.93$) and achieves a better—but still negative— R_{comm} . We call this failure

Table 4. Environment A at the converged budget (400 cycles; mean $\pm$ s.e.m.; $n=10$ seeds unless noted). “Closure” is the fraction of the oracle premium closed. Commit is the supporter’s commitment accuracy; R_{comm} on the information-gain scale.

Condition	R_{ext}	Closure	Commit	R_{comm}
Baseline	$0.845 \pm .010$	—	.851	—
Oracle	$0.957 \pm .008$	100%	.880	—
Simple L_0	$0.915 \pm .010$	62%	.857	-0.81
Exclusivity L_0 ($n=8$)	$0.795 \pm .081$	n.s.	.850	-1.65
Progress L_0	$0.924 \pm .008$	70%	.861	$+0.02$
Filter L_0	$0.920 \pm .006$	67%	.858	$+0.33$
Learned L_θ	$0.828 \pm .015$	-15%	.853	$+1.03$
Learned + RSA ($n=8$)	$0.863 \pm .008$	16%	.856	$+1.03$
Ear ($\lambda=0$, $n=6$)	$0.874 \pm .016$	26%	.854	—
Ear + R_{comm}	$0.878 \pm .012$	30%	.858	$+1.04$
Filter + ear	$0.920 \pm .012$	66%	.854	$+0.44$

mode miscalibrated signaling: a hand-crafted common ground rewards behavior the policy cannot realize and an observer cannot decode, converting the intrinsic reward into pure interference.

6.3.2. Learned Listeners: Calibrated Legibility

The learned listener changes the picture qualitatively. Both learned conditions steadily optimize their communicative reward from -0.65 toward zero (Fig. 10), confirming that the co-adapted reward is learnable—by construction, it only credits distinctions that are actually present in the behavioral distribution. This calibration eliminates the destructive interference: task performance is statistically indistinguishable from the baseline ($1.03 \pm .10$ and $1.09 \pm .04$ vs. $1.06 \pm .03$), with the pragmatic variant nominally best on task reward and specialization. During training, the co-adapting listener’s accuracy climbs from chance ($1/3$) to $\approx .43$ and is still rising at the end, with the RSA-recursed variant consistently ahead of the purely literal one—suggesting that contrastive reasoning over alternatives extracts signal that maximum-likelihood decoding alone does not. Specialization and probe accuracy do not differ measurably across conditions in this environment (consistent with its near-zero premium), so we omit their curves.

6.3.3. How Much Was There to Say? Oracles and Post-Hoc Probes

Two further measurements put the above in perspective. First, the oracle: giving agents their teammates’ true preferences does not improve task performance (0.99 vs.

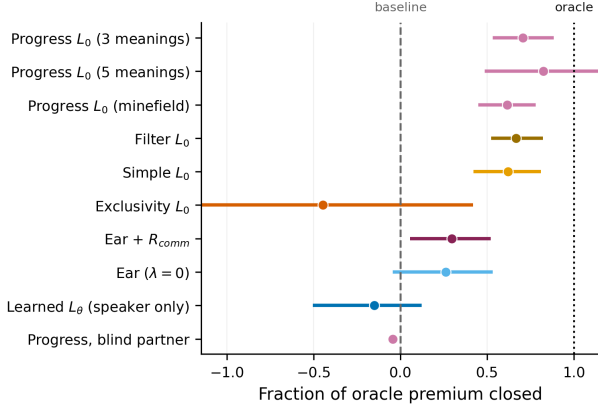


Fig. 3. The results on one axis: fraction of each environment’s oracle premium closed (dots; lines are bootstrap 95% CIs over seeds). Bounded motion-grounded listeners transfer across environments; the saturated learned listener and the blind control sit at zero.

Table 5. Final performance (mean $\pm$ s.e.m. over 3 seeds, averaged over the last three evaluations). R_{ext} is the per-step team reward; Spec. is team specialization; R_{comm} is each condition’s own communicative reward (information-gain scale); Post-hoc is the decoding accuracy of a fresh listener trained on the final policy’s rollouts (chance = .33).

Condition	R_{ext}	Spec.	R_{comm}	Post-hoc
Baseline	$1.06 \pm .03$	.847	—	$.598 \pm .02$
Oracle	$0.99 \pm .05$	.841	—	$.589 \pm .01$
Simple L_0	$0.93 \pm .07$	.833	-1.33	$.599 \pm .01$
Exclusivity L_0	$0.57 \pm .05$	.815	-2.39	$.604 \pm .02$
Learned L_θ	$1.03 \pm .10$	.847	-0.13	$.582 \pm .03$
Learned + RSA	$1.09 \pm .04$	.853	-0.13	$.578 \pm .02$

1.06). Near-optimal coordination in this environment is achievable from positions and velocities alone—who is closest to what resolves most conflicts—so the marginal task value of intent information is small. Second, the post-hoc probes: a fresh listener trained on any condition’s converged behavior decodes intentions at .58–.60 accuracy (Table 5), with no significant differences across conditions. Because acting on one’s preference is extrinsically rewarded, converged behavior is already informative; at $\lambda = 0.1$ there is simply little decodable information left for a calibrated reward to add—which the learned listener correctly reflects by saturating near zero, and which the hand-crafted listeners ignore by charging a permanent task tax for no informational return.

Together these results reframe the evaluation of implicit-communication methods: task reward alone

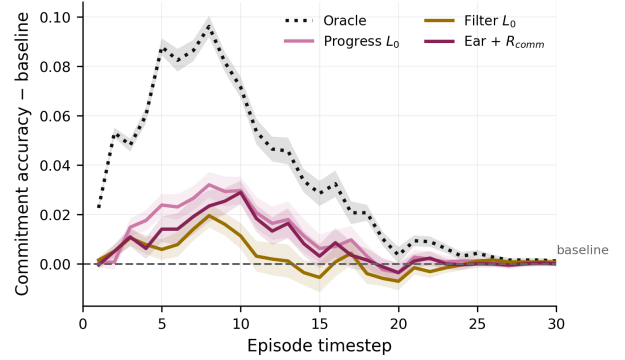


Fig. 4. Supporter commitment accuracy relative to baseline over the episode (converged policies, 256 held-out episodes; bands combine both groups’ s.e.m.). Plotting the difference makes the small, consistent timing gains visible where absolute curves overlap: motion-grounded listeners and the ear buy a fraction of the oracle’s early-commitment advantage during the decision window ($t \lesssim 20$).

conflates the cost of signaling with its value, and a method should be judged on (a) whether its intrinsic reward is achievable (calibration), (b) how much decodable information the resulting behavior carries (post-hoc probes), and (c) what it pays for that information (task-reward gap). On calibration and task cost the learned listener dominates the hand-crafted ones; on post-hoc decodability the conditions tie (.58–.60, n.s.), consistent with this environment’s near-zero premium.

6.3.4. The Weight λ : Dial or Landmine?

We sweep the communicative weight for both listener families in Environment B (2 seeds per new point; Fig. 7). With the learned listener, λ behaves like a dial along an information-performance frontier: $\lambda = 0.1$ is free ($R_{\text{ext}} = 1.03$, $R_{\text{comm}} = -0.13$); $\lambda = 0.3$ buys near-saturated clarity ($R_{\text{comm}} = -0.05$) for a modest task cost ($R_{\text{ext}} = 0.84$); and $\lambda = 1.0$ pushes the communicative reward to genuinely positive information gain ($R_{\text{comm}} = +0.62$)—behavior more informative than task-optimal motion—at a large task cost ($R_{\text{ext}} = -0.80$) as agents privilege signaling over covering landmarks. With the exclusivity listener, no operating point buys information: at $\lambda = 0.03$ the reward is simply ignored (R_{comm} stuck at -2.6, task performance at baseline level), and at $\lambda = 0.1$ it destroys 46% of task reward with R_{comm} still at -2.4. Calibration is precisely what turns λ from a landmine into a dial.

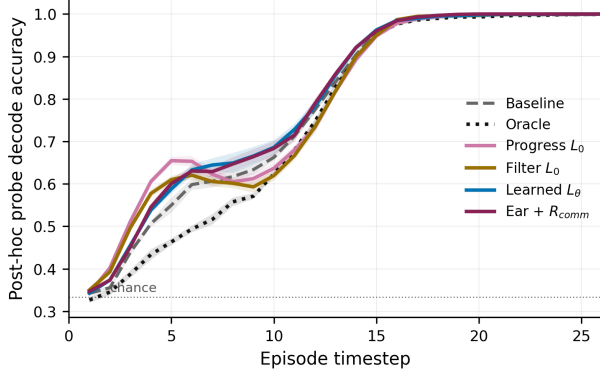


Fig. 5. Information profile: per-timestep decode accuracy of a fresh post-hoc probe on each condition’s converged behavior (first 26 steps shown; all saturate by $t \approx 16$). During the early decision window ($t \lesssim 10$) the progress-listener scouts carry the most decodable intent, followed by the filter and ear conditions, while oracle scouts—whose teams never rely on motion—are the least legible early. This is the mechanism behind the recovered gap: legibility is bought exactly when the partner is still deciding.

6.3.5. Qualitative Behavior

Figure 8 shows a representative held-out episode of Environment B. The baseline reaches landmarks but meanders, with late target switches. The exclusivity-listener agent displays the miscalibration diagnosed above: trajectories sweep across the arena and settle far from any landmark—costly motion signaling nothing. The learned-pragmatic agent produces direct, mutually distinct paths: each agent claims a separate landmark, and the agent whose preferred landmark is taken by a teammate cleanly re-targets the remaining one.

7. DISCUSSION

The common ground is the problem. Our central empirical finding is that the success of a pragmatic reward hinges almost entirely on the alignment between the literal listener and the agents’ realizable behavior. A hand-crafted L_0 encodes the designer’s theory of what signaling should look like; when that theory is even mildly wrong—as with the exclusivity margin, which demands geometrically unsatisfiable action patterns—the recursion built on top of it faithfully amplifies the error, and the agent is paid to produce elaborate, undecodable gestures. This failure is quiet: behavioral metrics (path deviation, action magnitude, early probe accuracy) suggest the agent is “signaling harder,” while the communicative channel carries nothing. We suspect this failure mode is

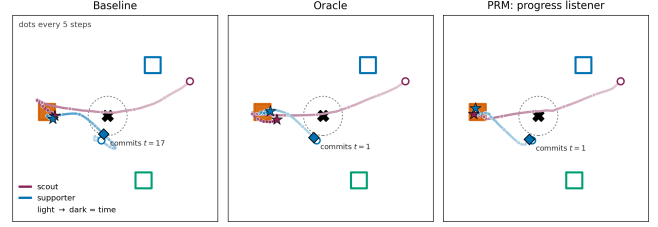


Fig. 6. The same Environment A evaluation episode under three conditions (converged policies, seed 0; the episode was selected, from 60 candidates, to maximize the baseline-vs-recipe commitment-time difference—it illustrates the mechanism, the statistics are in Table 4). Squares are sites (the true target filled); the black cross and dashed circle mark the pickup waypoint the scout must visit first. Trajectories darken with time (dot every 5 steps; dense dots = slow motion); open circle = start, star = end, diamond = the supporter’s commitment moment. The baseline supporter hesitates mid-arena (stacked dots) and commits only at $t=17$; the pragmatically rewarded supporter commits at $t=1$ and drives straight to the site, matching the oracle’s behavior—early commitment bought by the scout’s legible first leg.

widespread in reward designs that score behavior against idealized observer models, and recommend a simple diagnostic that requires no extra machinery: track whether the agent can drive its own intrinsic reward above the uninformative level. A persistently negative information gain is a certificate of miscalibration.

Learning the listener closes the loop—at a price. Training L_θ on the agents’ own replay data guarantees that the common ground never demands what the policy distribution does not contain; the reward then only sharpens distinctions that already exist, which is why it optimizes smoothly and costs no task performance where there is nothing to say (Environment B). Proposition 1 explains the mechanism: the learned-listener reward is the tightest variational mutual-information bound available from the data, so policy improvement against it is information maximization—grounded, by construction, in a decodable channel. But the same property is the seed of the saturation failure where there is something to say: a co-adapting maximum-likelihood listener converges toward the true posterior, and the closer it gets, the less its log-posterior varies with behavior (Proposition 2). Calibration and shaping pressure trade off against each other. The RSA recursion partially escapes this bind: by scoring the chosen action against foregone alternatives it re-introduces contrast that the converged posterior no longer supplies, which is where its consistent empirical contribution comes from (Sec. 6.1.1).

Communication is a two-sided market—and the sides

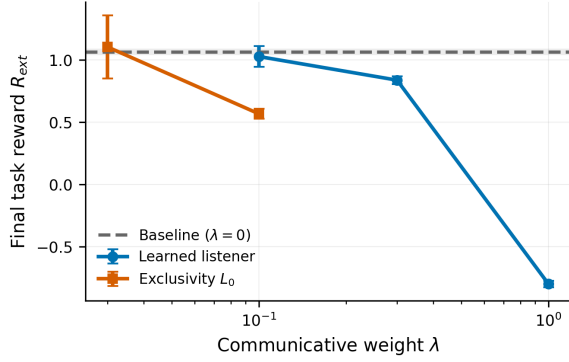


Fig. 7. Final task reward vs. communicative weight λ . The learned listener trades task reward for information smoothly; the exclusivity listener offers no λ at which information is gained.

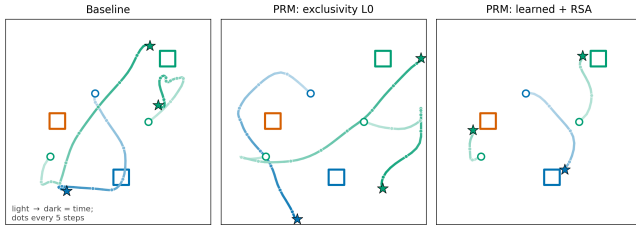


Fig. 8. A held-out episode under three conditions. Squares are landmarks (color = category); each trajectory is colored by the agent’s private preference and darkens with time, with a dot every 5 steps, so dense dots mean slow or stationary motion. Open circle = start, star = end.

can substitute. Environment A decomposes implicit communication into speaker and receiver contributions, and the converged results show two distinct routes across the gap. The overt-signal route shapes the speaker with a bounded listener until its motion is readable by the teammate’s ordinary perception; here all the machinery is on the speaker side, and the receiver needs nothing but its own policy network. The equipped-receiver route leaves the speaker’s task-driven behavior untouched and invests in decoding, deploying the training listener at execution. Both are decentralized-executable; in our benchmark the overt-signal route recovers about twice as much of the oracle premium, suggesting that when the intended reader is weak—an RL policy, plausibly a human—moving the burden of clarity onto the speaker is the more effective allocation. The joint-exploration pathology of emergent communication [15] appears here only transiently, as a mid-training phenomenon.

The audience-matching principle, sharpened by its own ablation. Our strongest finding is an inversion that

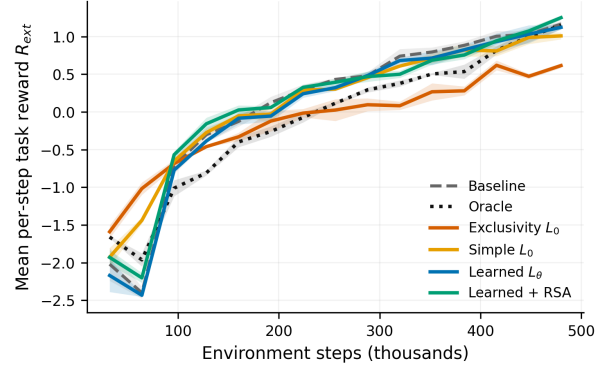


Fig. 9. Mean per-step team reward R_{ext} (3 seeds, mean $\pm$ s.e.m.).

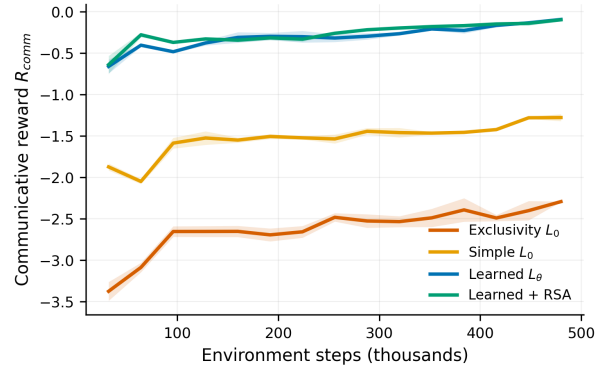


Fig. 10. Achieved communicative reward R_{comm} (each condition under its own listener; information-gain scale, 0 = uninformative).

computational pragmatics anticipated—and our attempt to exploit it constructively is what gives the principle its final form. The learned listener, trained to convergence, decodes the scout from raw observations and motion history at 97% regardless of how the scout behaves: its reward saturates, and by Proposition 2 the pragmatic gradient vanishes. The obvious remedy—bound the listener to the audience’s own viewpoint and decision window—turns out to be necessary but not sufficient (Sec. 6.1.1): a bounded listener that co-adapts still saturates, by chasing whatever code the current policy emits, and its literal reward still shapes nothing. What separates the crude geometric listeners is therefore not weakness per se but what they are able to credit: only overtly geometric motion—and overt motion is exactly what a feed-forward teammate reading raw observations can also exploit. An omniscient audience needs no speech; an infinitely accommodating one accepts any private code. The working form of the audience-matching principle is thus two-sided: anchor R_{comm} in a listener that (a) is no

stronger than the weakest intended reader and (b) credits only the signal vocabulary that reader actually uses. The one component that survives learning is RSA’s recursion itself: at a raised weight, contrastive reasoning over alternatives converts an inert decoder into a reward that recovers a third of the premium, while the same decoder without recursion stays below baseline at the same weight—pragmatics helps precisely where decoding alone is inert, because the converged posterior no longer varies with behavior but the contrast against foregone alternatives still does.

Clarity must be paid when it is needed—during learning. A second, dynamical lesson: mid-training, rewards that concentrate clarity where the partner’s decisions hang (our value-of-information weighting) visibly steer emergence, while uniformly-integrated clarity is spent on behavior that is already informative. At convergence these weighting effects wash out, but the trajectory through which a team reaches its equilibrium is shaped by when clarity is subsidized—a distinction invisible to asymptotic evaluation alone.

When does legibility pay? The contrast between environments is a caution for the field: in Environment B, where geometry resolves coordination, even a perfect oracle gains nothing, and a well-calibrated legibility reward correctly approaches a no-op; in Environment A the same machinery faces a certified premium and recovers most of it. Benchmarks for communicative MARL should report their oracle gap—without it, task reward conflates the cost of signaling with its value. Where the gap is large, our results argue for evaluating methods on three separable axes: calibration of the intrinsic reward, information actually transmitted when the partner needs it (time-resolved probes), and the task price paid for it—all measured at, and reported across, converged training budgets.

7.1. Limitations and Future Work

Several limitations qualify our findings and chart future work. First, the intrinsic reward densifies the learning signal, which can aid learning independently of legibility; our commitment-timing and probe metrics mitigate but do not eliminate this confound. Second, the audience-matching principle is demonstrated for one audience class (feed-forward policies over raw observations); its quantitative form—how listener competence should scale with reader competence—is open, and the wide-model check suggests the premium itself shrinks as readers strengthen. Third, meanings are hand-specified discrete variables; extending R_{comm} to continuous, hierarchical, or self-assigned meaning spaces—where the agent learns what is worth communicating, not only how—remains open. Fourth, validation is confined to AI-AI teams

in 2D particle worlds; the audience-matching principle makes a testable prediction for human-agent teams—rewards anchored in human-competence listeners (e.g., L_θ fit to human intent judgments) should transfer better than either extreme—which we view as the natural next experiment. Finally, the framework is symmetric in an intriguing way: inverting the sign of λ turns legibility into obfuscation, suggesting privacy-preserving applications where agents remain opaque to opponents while staying legible to allies [4].

8. CONCLUSION

We presented a framework that connects pragmatic inference and maximum-entropy reinforcement learning, importing the RSA listener value as a principled intrinsic reward for legible behavior, with the literal listener—the assumed common ground—as its decisive design choice, and learned listeners yielding the tightest of a family of information bounds that every listener in the framework shares. Our study then locates where the choice bites. Where communication has no task value, miscalibrated listeners charge for undecodable signaling while calibrated ones correctly approach a no-op. Where communication is certifiably valuable, crude motion-grounded listeners, which only credit overt signals, recover 60–82% of the oracle premium through the speaker-side reward alone, while a listener competent enough to decode anything demands nothing and shapes nothing—a failure we prove (gradient saturation), then probe constructively: bounding the learned listener to the audience’s viewpoint and decision window cures its harm but not its inertness, RSA’s contrastive recursion at a raised weight recovers a third of the premium where the weight-matched literal control recovers none, and the hand-crafted restriction to the audience’s own signal vocabulary remains the strongest reward. The audience-matching principle that emerges is, we believe, RSA’s oldest lesson restated for reward design: speech is optimized against a literal listener precisely because the literal listener is limited—in competence, and in what it is willing to hear. Agents whose competence includes being understood must be trained against audiences that need the help.

A. HYPERPARAMETERS

Table 6. Hyperparameters (all conditions unless noted).

Parameter	Value
Parallel environments	64
Episode length	50
Training cycles (Env. B / Env. A)	150 / 400
Actor / critic architecture	MLP 256×256
Learning rate (actor, critic)	3×10^{-4}
Discount γ	0.95
Target smoothing τ	0.005
Batch size / updates per env step	512 / 4
Entropy target	$-\dim(\mathcal{A})$ (auto-tuned)
Listener L_θ	MLP $128 \times 128 \times 3$
Direction listener temp. $\alpha_{\text{temp}} / K / N_r$	5 / 64 / 1
Learned+RSA alternatives / N_r	16 / 1
Filter listener temp. / forgetting ρ	2 / 0.9
λ (Env. B / Env. A)	0.1 / 0.3
VOI post-key weight (Env. A)	0.2
Scout / supporter max speed (Env. A)	1.0 / 0.6
Scout start annulus radius (Env. A)	[0.9, 1.3]
Bonus / cover / pickup radius (Env. A)	3.0 / 0.3 / 0.3
Motion-history window (Env. A)	6 steps
Seeds (headline / other conditions)	10 / 3–8
Integration step / damping / accel. gain	0.1 / 0.25 / 5.0
Arena clamp / site jitter / waypoint jitter	± 1.5 / 0.1 / ± 0.25
Mine penalty (minefield variant)	1.5
Replay capacity / random warm-up	400k / 2000 steps
Evaluation	every 10 cycles, fixed 64 episodes
Final estimate	mean of last 3 evaluations

Reproducibility. All environments, training, aggregation, and figure code, together with the launch specifications (JSON files that determine every run in this paper exactly, including seeds), will be released with the camera-ready version.

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